

Acoustic gravitational waves from primordial curvature perturbations

Zhuan Ning (宁专), UCAS

Based on: *Sound waves from primordial black hole formations*, arXiv: 2504.12243

Relic gravitational waves from primordial gravitational collapses, arXiv: 2504.11275

Acoustic gravitational waves from primordial curvature perturbations, arXiv: 2512.21151

Co-authors: Xiang-Xi Zeng, Zi-Yan Yuwen, Shao-Jiang Wang, Heling Deng, Rong-Gen Cai

At Tsinghua University

2025.12.27



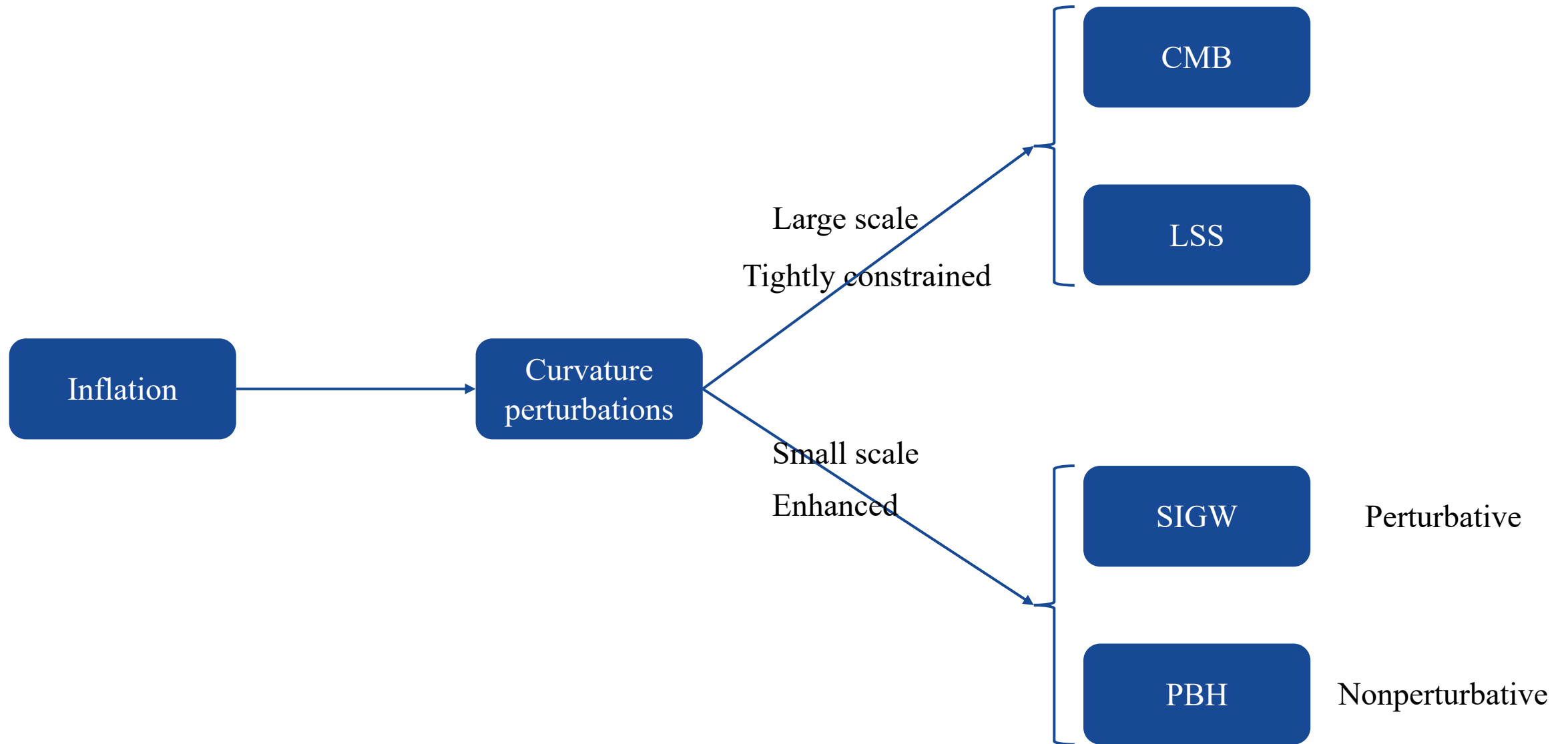
Outline

- Introduction
- Sound waves from collapsing curvature peaks
- Sound shell model
- 3D simulations of acoustic gravitational waves
- Numerical results
- Conclusions and discussion

Outline

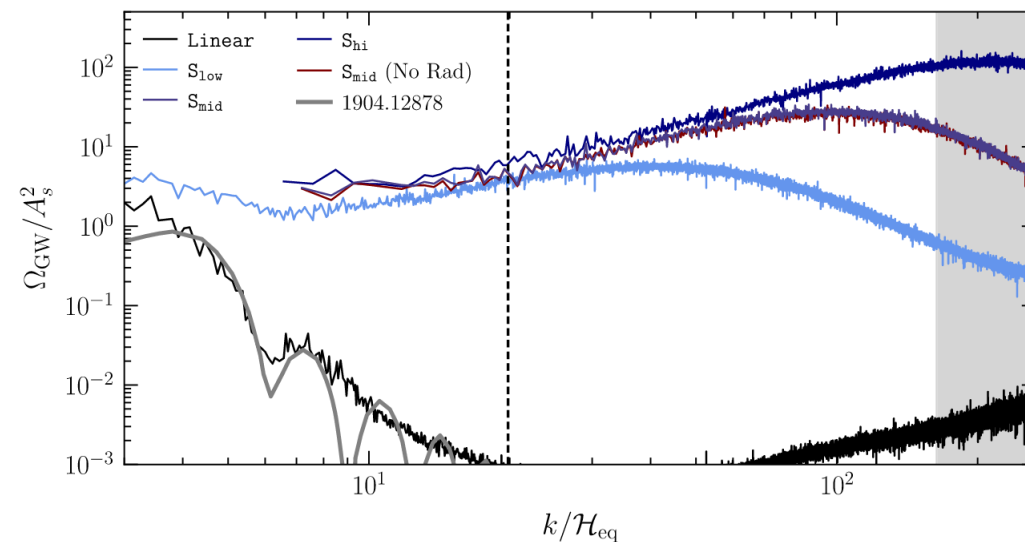
- Introduction
- Sound waves from collapsing curvature peaks
- Sound shell model
- 3D simulations of acoustic gravitational waves
- Numerical results
- Conclusions and discussion

Introduction



Introduction

- SIGWs omit some nonperturbative effects!
- **Non-spherical collapse** of peaks in curvature perturbations [[Luca et al. 1905.13459](#)]
- Early matter-dominated era [[0901.0989](#), [1002.3278](#), [2212.00425](#), [2312.12499](#)]
 - Collapse of halos
 - Tidal interactions
 - Evaporation of halos into radiation
 - Dissipation after reheating
- **Amplified sound waves**



Sound waves from primordial black hole formations, [arXiv: 2504.12243](#)

Relic gravitational waves from primordial gravitational collapses, [arXiv: 2504.11275](#)

Acoustic gravitational waves from primordial curvature perturbations, [arXiv: 2512.21151](#)

Outline

- Introduction
- Sound waves from collapsing curvature peaks
- Sound shell model
- 3D simulations of acoustic gravitational waves
- Numerical results
- Conclusions and discussion

Fully GR simulations

➤ Misner-Sharp formalism

- Metric ansatz: $ds^2 = -A(t, r)^2 dt^2 + B(t, r)^2 dr^2 + R(t, r)^2 d\Omega^2$
- Energy-momentum tensor: $T_{\mu\nu} = (\rho + p)u_\mu u_\nu + pg_{\mu\nu}$
- Equation of state: $p = \omega\rho$
- Initial conditions: **long-wavelength approximation**

$$ds^2 = -dt^2 + a(t)^2 e^{2\zeta(r)} (dr^2 + r^2 d\Omega^2)$$

$$\zeta(r) = \mu e^{-(r/r_m)^2}$$

$$\mu_c \approx 0.79579 \pm 1 \times 10^{-5}$$

$$D_r A = \frac{-A}{\rho + p} D_r p,$$

$$D_r M = 4\pi\Gamma\rho R^2,$$

$$\Gamma = \sqrt{1 + U^2 - \frac{2M}{R}},$$

$$D_t R = U,$$

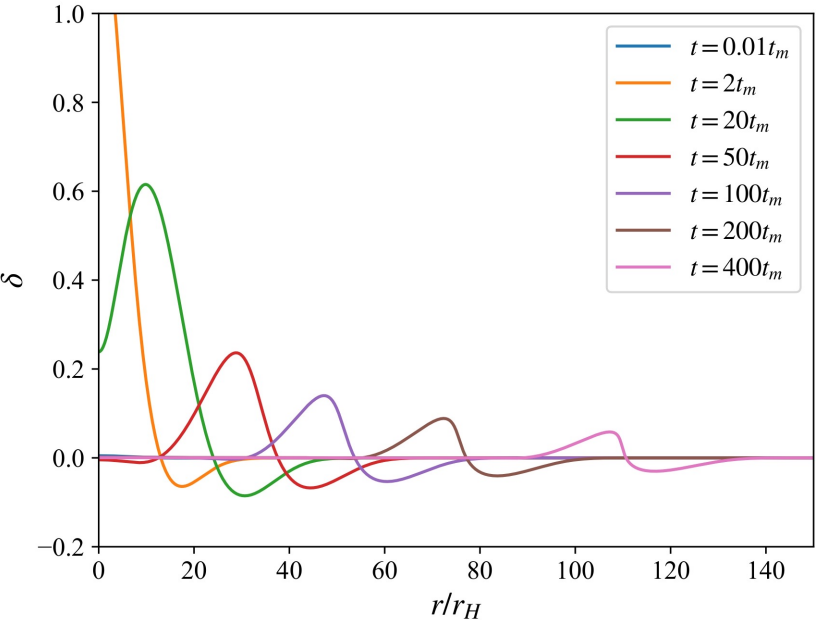
$$D_t U = - \left[\frac{\Gamma}{(\rho + p)} D_r p + \frac{M}{R^2} + 4\pi R p \right],$$

$$D_t M = -4\pi R^2 U p,$$

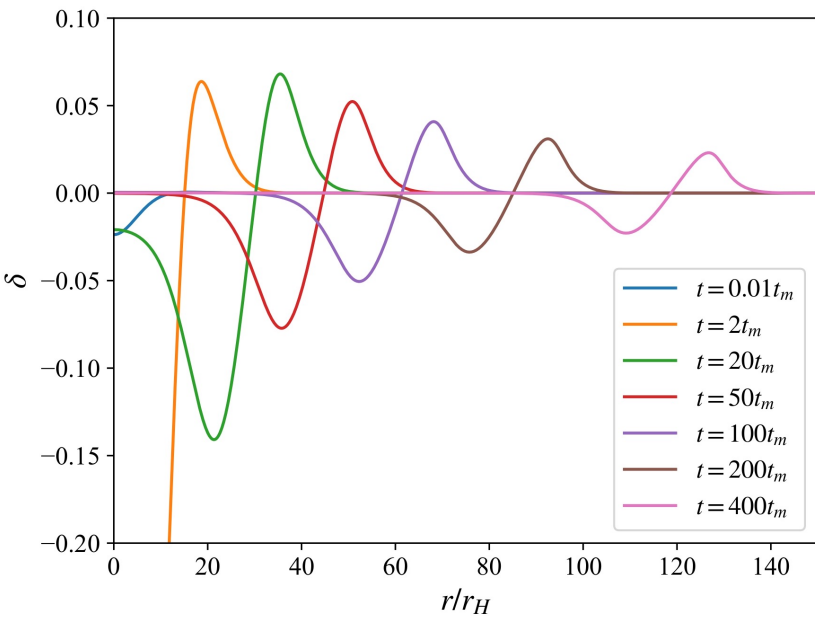
$$D_t \rho = - \frac{(\rho + p)}{\Gamma R^2} D_r (U R^2).$$

Fully GR simulations

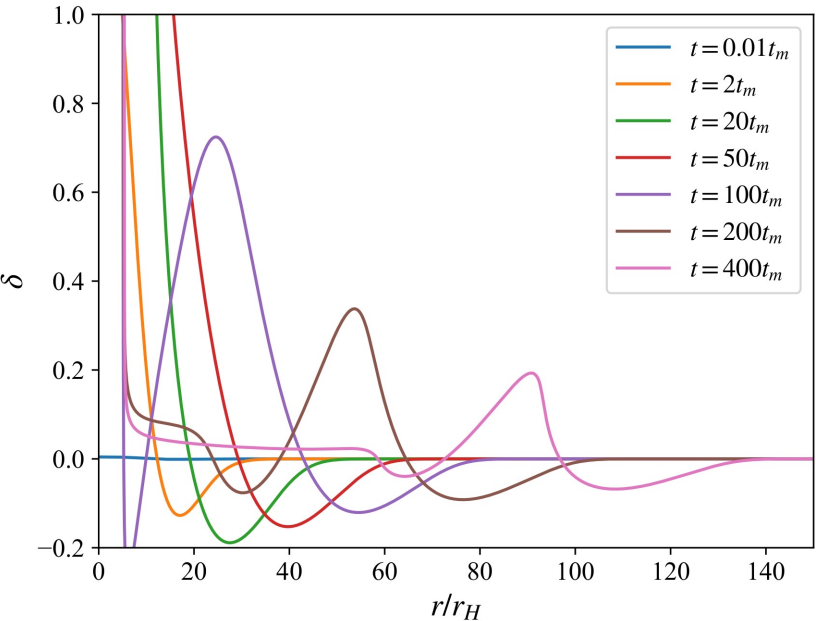
subcritical
($\mu = 0.4$)



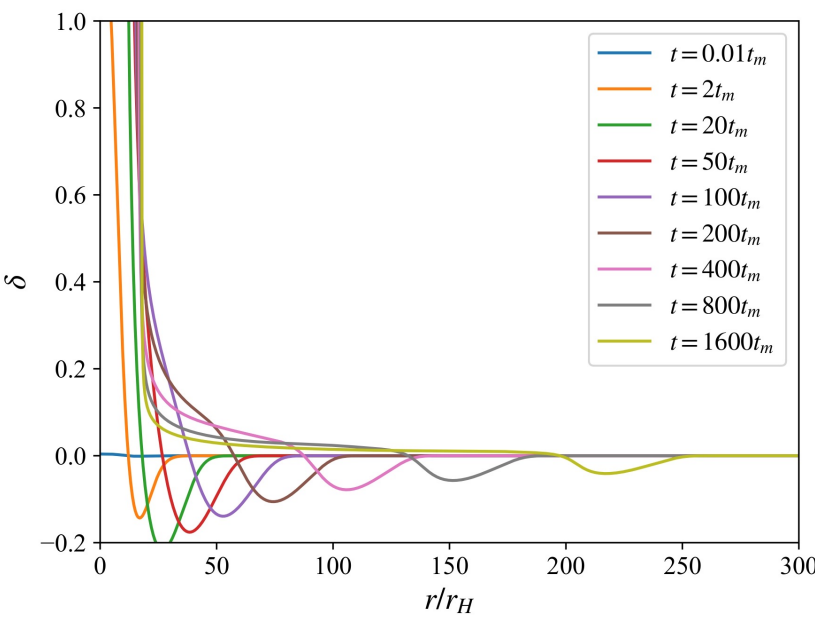
negative
($\mu = -0.4$)



near-critical
($\mu = 0.8$)



supercritical
($\mu = 0.9$)



Comparison with linear evolution

➤ Cosmological perturbation theory

- Metric ansatz: $ds^2 = a(\eta)^2 [-(1 + 2\phi)d\eta^2 + 2\partial_i\beta d\eta dx^i + [(1 - 2\psi)\delta_{ij} + 2\partial_i\partial_j E] dx^i dx^j]$
- Comoving orthogonal gauge ($\beta = v = 0$):

$$\ddot{\psi} + \left(\frac{3(3 + \omega)}{2} H + \frac{\dot{H}}{H} \right) \dot{\psi} - \frac{\omega}{a^2} \nabla^2 \psi = 0,$$
$$\phi = -\frac{\dot{\psi}}{H}, \quad \frac{\delta\rho}{\bar{\rho}} = -\frac{1 + \omega}{\omega} \phi,$$

- Initial conditions:

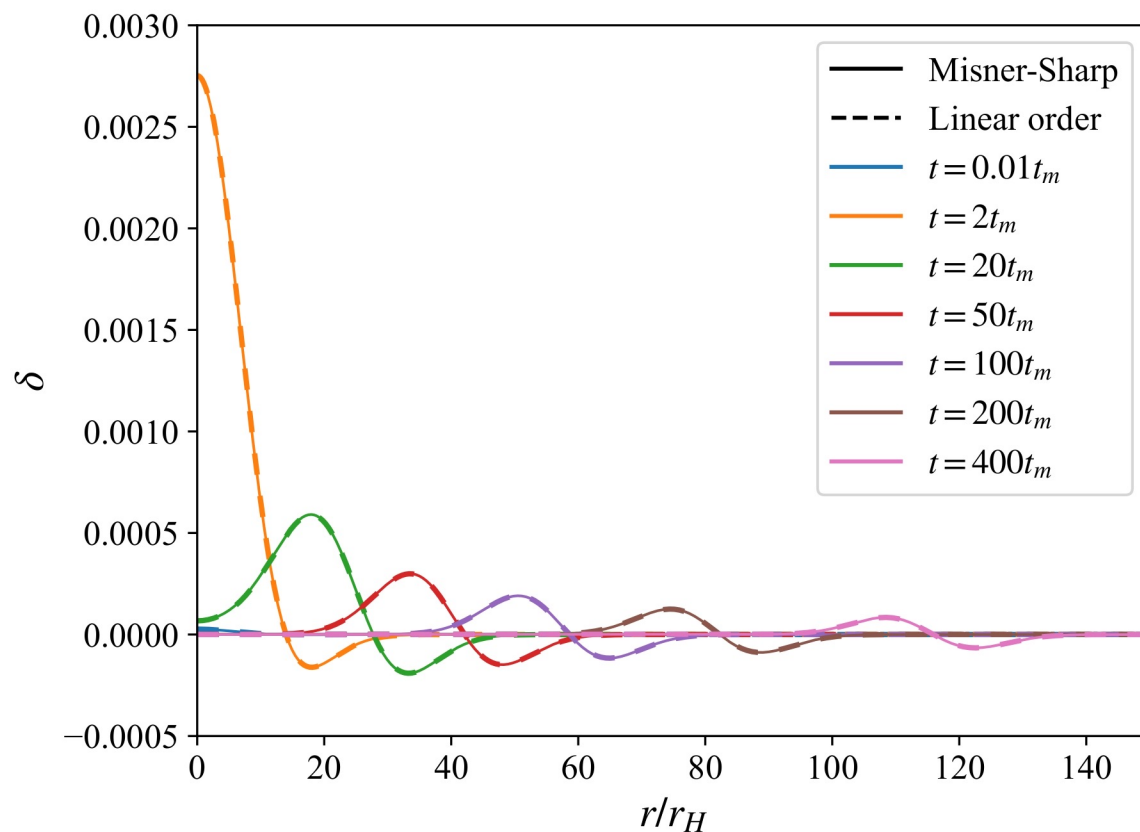
$$\zeta \equiv -\psi + \mathcal{H}(v + \beta) = -\psi$$

$$\phi = (A^2 - 1)/2$$

Comparison with linear evolution

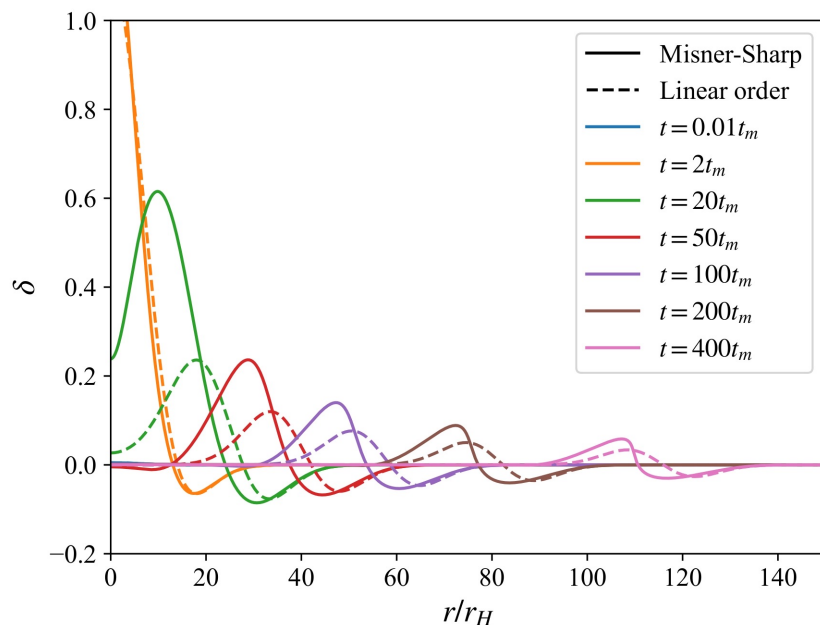
- When perturbations are small, the **fully nonperturbative Misner-Sharp formalism** should reproduce the results of linear perturbative evolution

$$\mu = 0.001$$

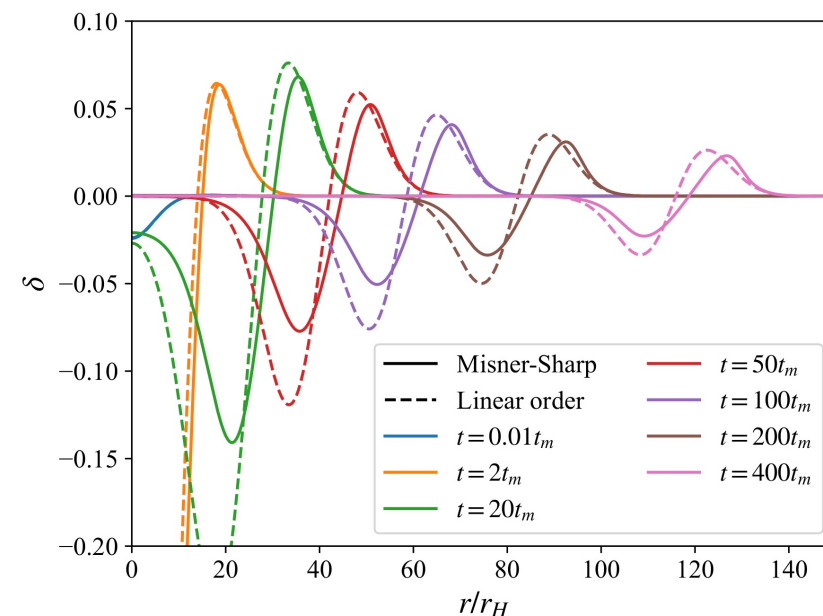


Comparison with linear evolution

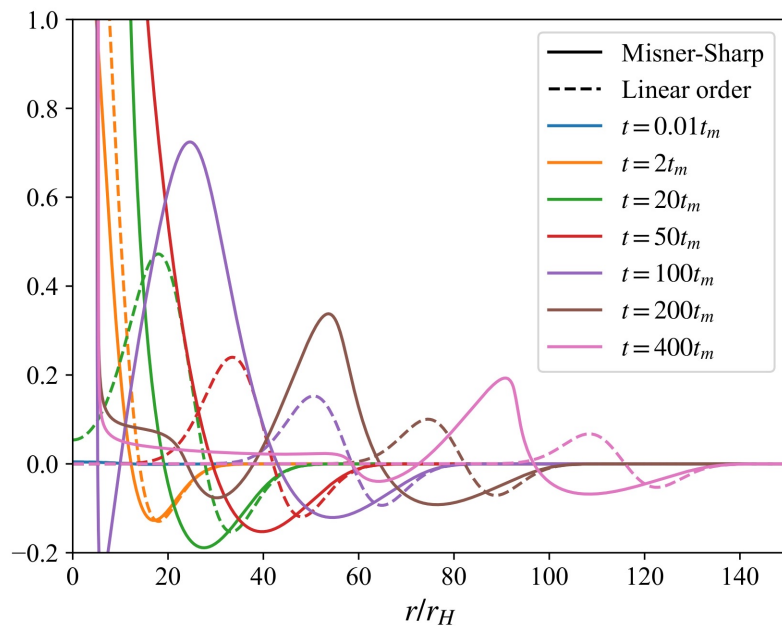
subcritical
($\mu = 0.4$)



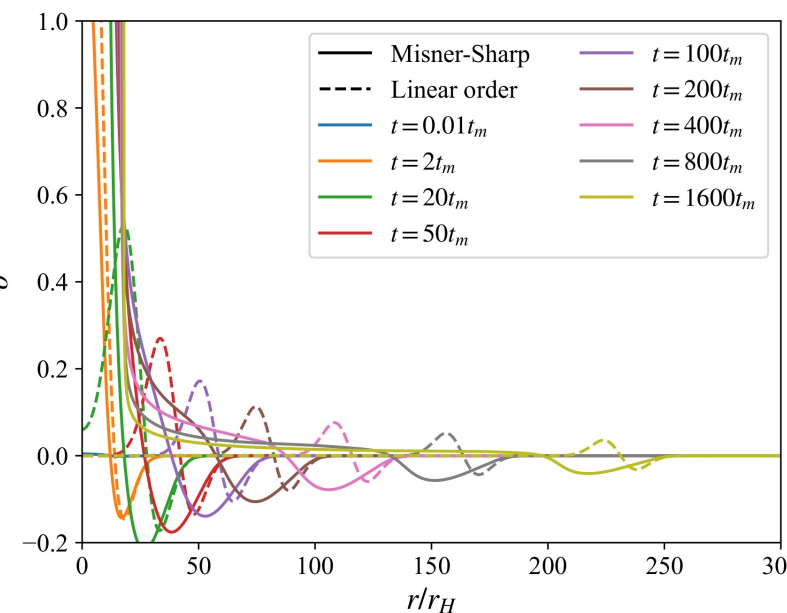
negative
($\mu = -0.4$)



near-critical
($\mu = 0.8$)



supercritical
($\mu = 0.9$)



Outline

- Introduction
- Sound waves from collapsing curvature peaks
- Sound shell model**
- 3D simulations of acoustic gravitational waves
- Numerical results
- Conclusions and discussion

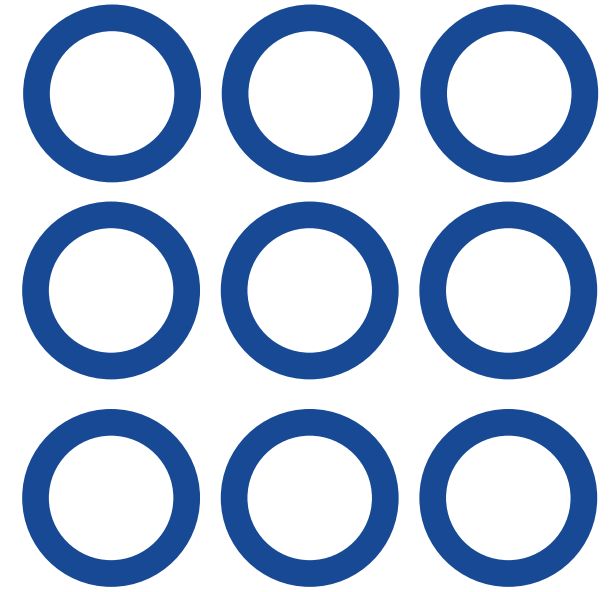
Sound shell model

➤ A semi-analytical method adopted from FOPTs with approximations:

- **Linear superposition** of sound shells
- **Uniform distribution** of peaks
- **Same amplitude** of peaks

$$\mathcal{P}_{\text{GW}}(t, k) \equiv \frac{d\Omega_{\text{GW}}(t)}{d \ln k}$$

$$\Omega_{\text{GW}}(t) \equiv \frac{\rho_{\text{GW}}(t)}{\rho_c(t)}$$



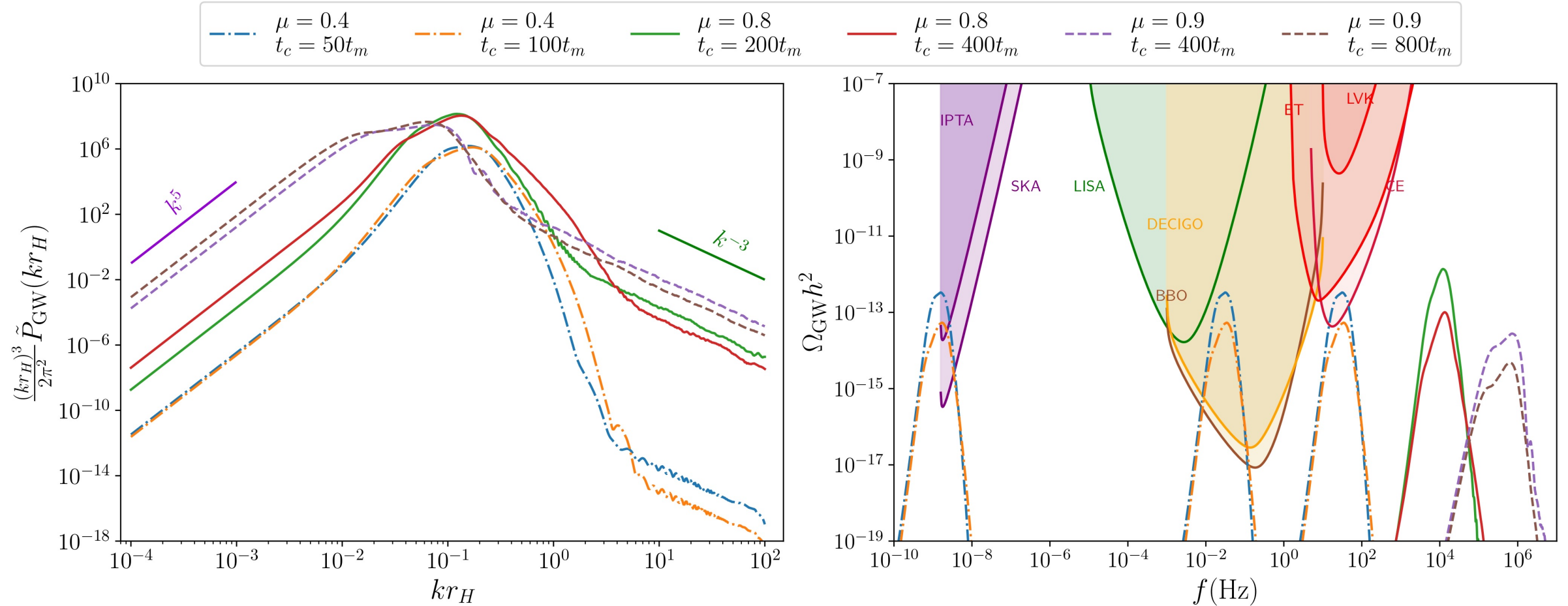
$$\mathcal{P}_{\text{GW}}(y, kr_H) = 3\Gamma^2 \underbrace{(H_s a_s R_{*c})}_{\text{const}} \underbrace{\left(\frac{r_H}{R_{*c}}\right)^7}_{\text{convolution of the velocity power spectrum}} \frac{(kr_H)^3}{2\pi^2} \times \tilde{P}_{\text{GW}}(kr_H) \Upsilon(y)$$

$(\bar{\rho} + \bar{p})/\bar{\rho} \approx 4/3$
1

R_{*c} : mean comoving separation between sound shells (peaks)

Sound shell model

➤ The dimensionless GW power spectrum

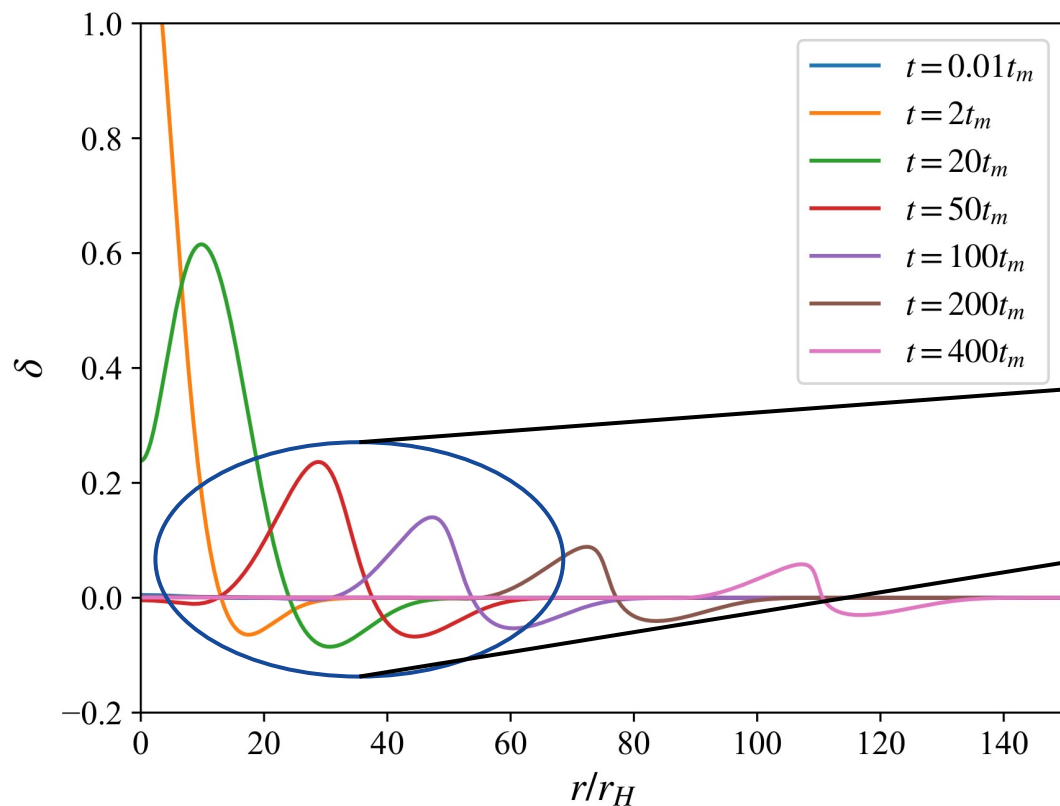


Outline

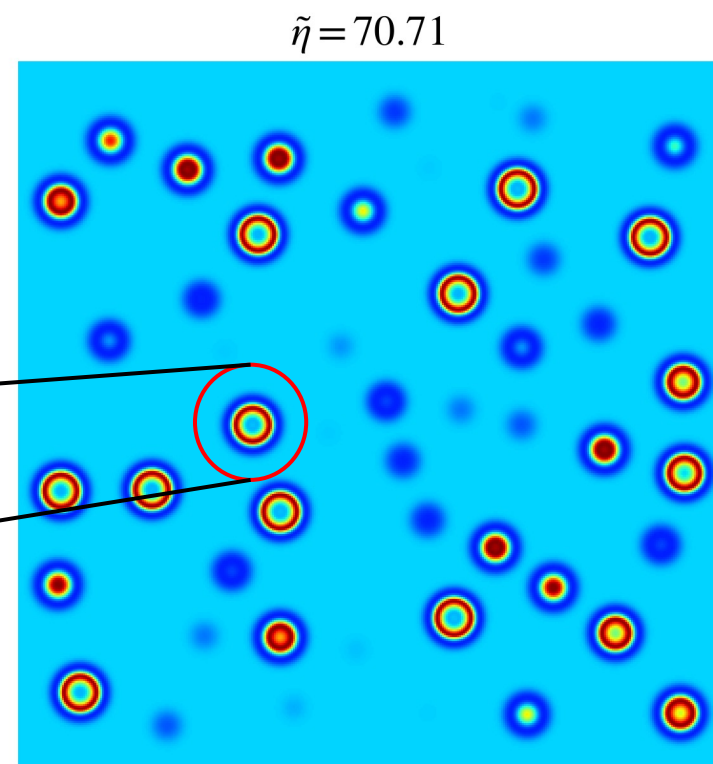
- Introduction
- Sound waves from collapsing curvature peaks
- Sound shell model
- 3D simulations of acoustic gravitational waves**
- Numerical results
- Conclusions and discussion

Hybrid simulations

- First stage: **gravitational collapse** of peaks to generate sound waves
- Second stage: **collisions between sound shells** to produce GWs
- An **end-to-end**, fully GR simulations is numerically prohibitive



Fully GR, spherically symmetric



Neglect other metric perturbations, 3D

Hybrid simulations

- Length scale of the curvature peaks: $r_m = 10r_H$ (initial Hubble horizon of 1D simulation)
- **Peak-like** power spectrum
- Sound shells are formed at the **same time**

Equations of motion

$$ds^2 = -a(\eta)^{2\alpha} d\eta^2 + a(\eta)^2 (\delta_{ij} + h_{ij}) dx^i dx^j$$

η : α -time (`CosmoLattice`). $\alpha = 0$: cosmic time, $\alpha = 1$: **conformal time**

➤ Relativistic fluid:

$$T^{\mu\nu} = (\rho + p)u^\mu u^\nu + pg^{\mu\nu}$$

$$u^0 = \gamma/a^\alpha, \quad u^i = \gamma v^i/a$$

$$\gamma = \frac{1}{\sqrt{1 - v^2}} = \frac{1}{\sqrt{1 - \delta_{ij} v^i v^j}}$$

$$\hat{T}^{00} = a^{2\alpha+4} T^{00},$$

$$\hat{T}^{0i} = a^{\alpha+5} T^{0i},$$

$$\hat{T}^{ij} = a^6 T^{ij}.$$

$$\partial_\eta \hat{T}^{00} + a^{\alpha-1} \partial_i \hat{T}^{0i} + a^4 \mathcal{H} (3p - \rho) = 0,$$

$$\partial_\eta \hat{T}^{0i} + a^{\alpha-1} \partial_j \hat{T}^{ij} = 0.$$

$$\hat{T}^{ij} = \frac{\hat{T}^{0i} \hat{T}^{0j}}{\hat{T}^{00} + a^4 p} + a^4 p \delta^{ij}$$

Equations of motion

➤ Spacetime:

$$\mathcal{H}^2 = \left(\frac{a'}{a}\right)^2 = \frac{a^{2\alpha}}{3M_{\text{Pl}}^2} \langle E \rangle,$$

$$\frac{a''}{a} = \frac{a^{2\alpha}}{6M_{\text{Pl}}^2} \langle (2\alpha - 1)E - 3P \rangle,$$

$$E = a^{2\alpha} T^{00} = \hat{T}^{00}/a^4,$$

$$P = a^2 \sum_i T^{ii}/3 = \sum_i \hat{T}^{ii}/(3a^4).$$

➤ GW:

$$u''_{ij} + (3 - \alpha)\mathcal{H}u'_{ij} - a^{2\alpha-2}\nabla^2 u_{ij} = 2a^{2\alpha-2}\Pi_{ij}^{\text{eff}}$$

$$\Pi_{ij}^{\text{eff}} = a^2(\rho + p)\gamma^2 v_i v_j = a^2(\rho + p)\gamma^2 v^i v^j = (\hat{T}^{ij} - a^4 p \delta^{ij})/a^2$$

$$\tilde{\eta} = \omega_* \eta, \quad \tilde{x} = \omega_* x, \quad \tilde{T}_{\mu\nu} = \frac{\hat{T}_{\mu\nu}}{f_*^2 \omega_*^2}, \quad \tilde{u}_{ij} = \left(\frac{M_{\text{Pl}}}{f_*}\right)^2 u_{ij}$$

$$\omega_* = 1/(a_1 r_H) \text{ and } f_* = 1$$

Representative slices

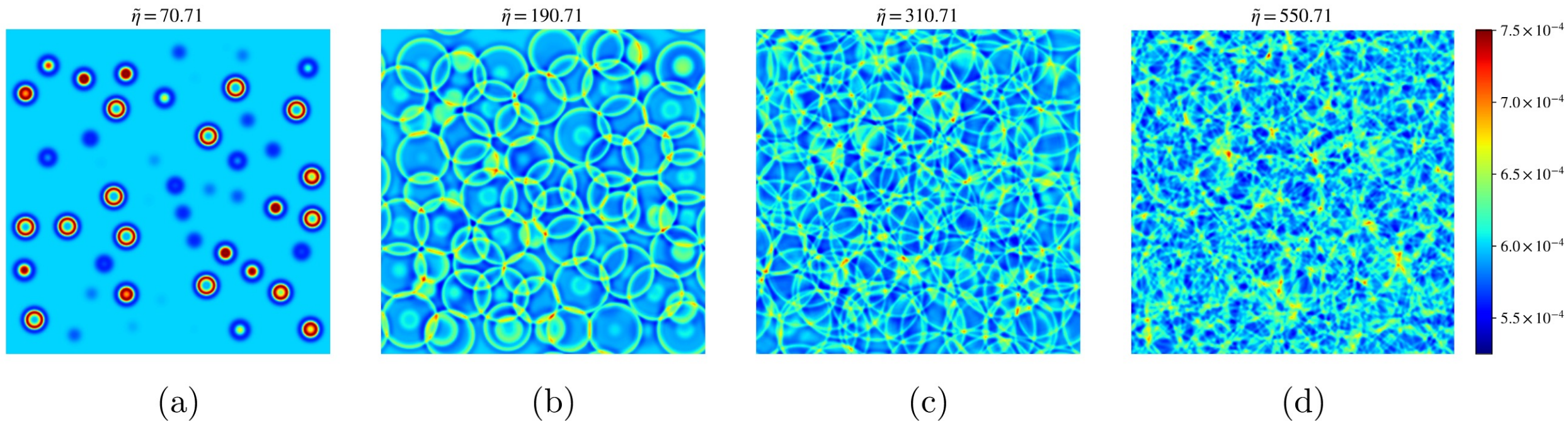


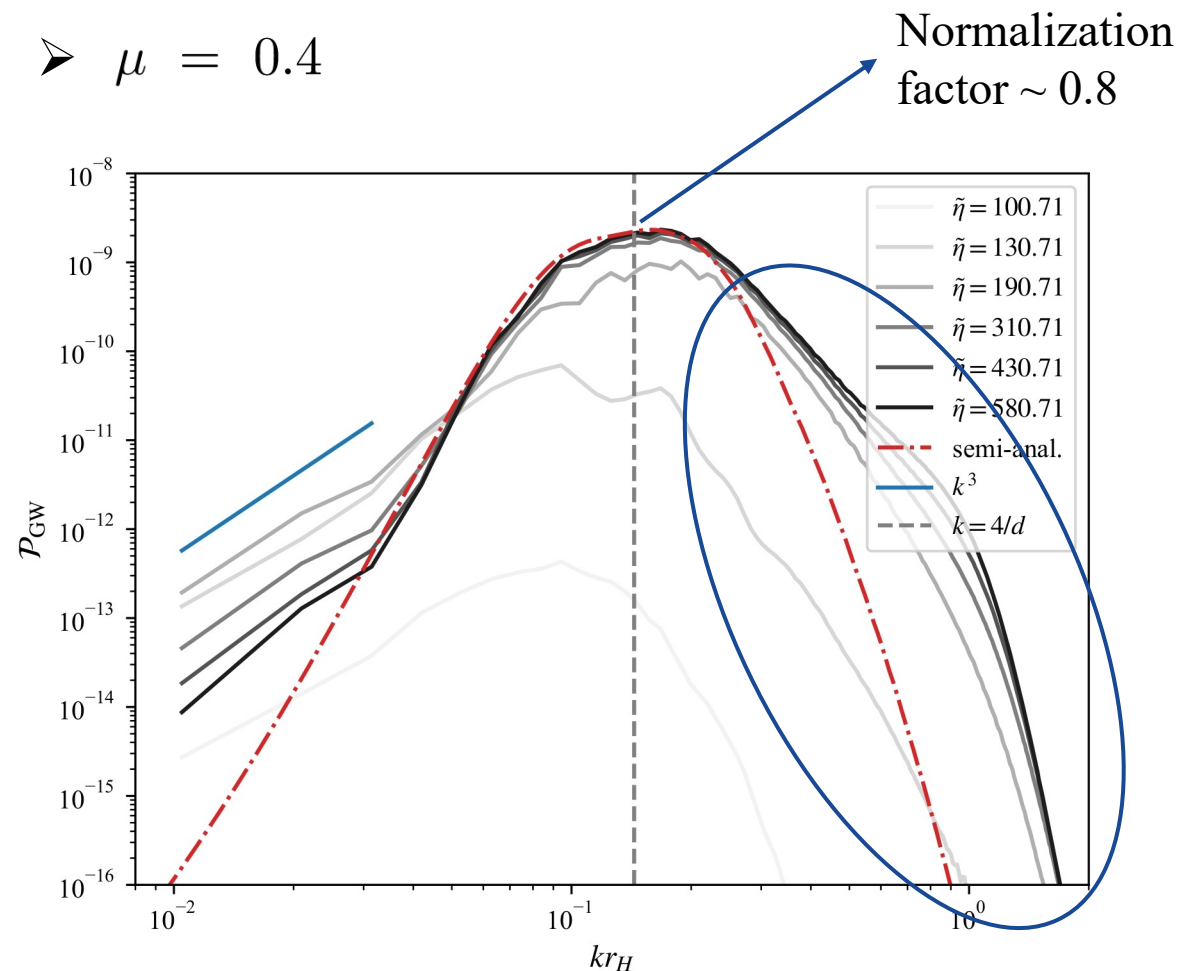
Figure 3: Slices of $\tilde{T}^{00}$ from a representative simulation. The initial condition is constructed by embedding the 1D sound-shell profile with $\mu = 0.4$ at $t_i = 50t_m$; the box length is $\tilde{L} = 1200$, and the number of sound shells is $N_s = 400$. The four snapshots correspond to $\tilde{\eta} = 70.71$ (a), $\tilde{\eta} = 190.71$ (b), $\tilde{\eta} = 310.71$ (c), and $\tilde{\eta} = 550.71$ (d).

Outline

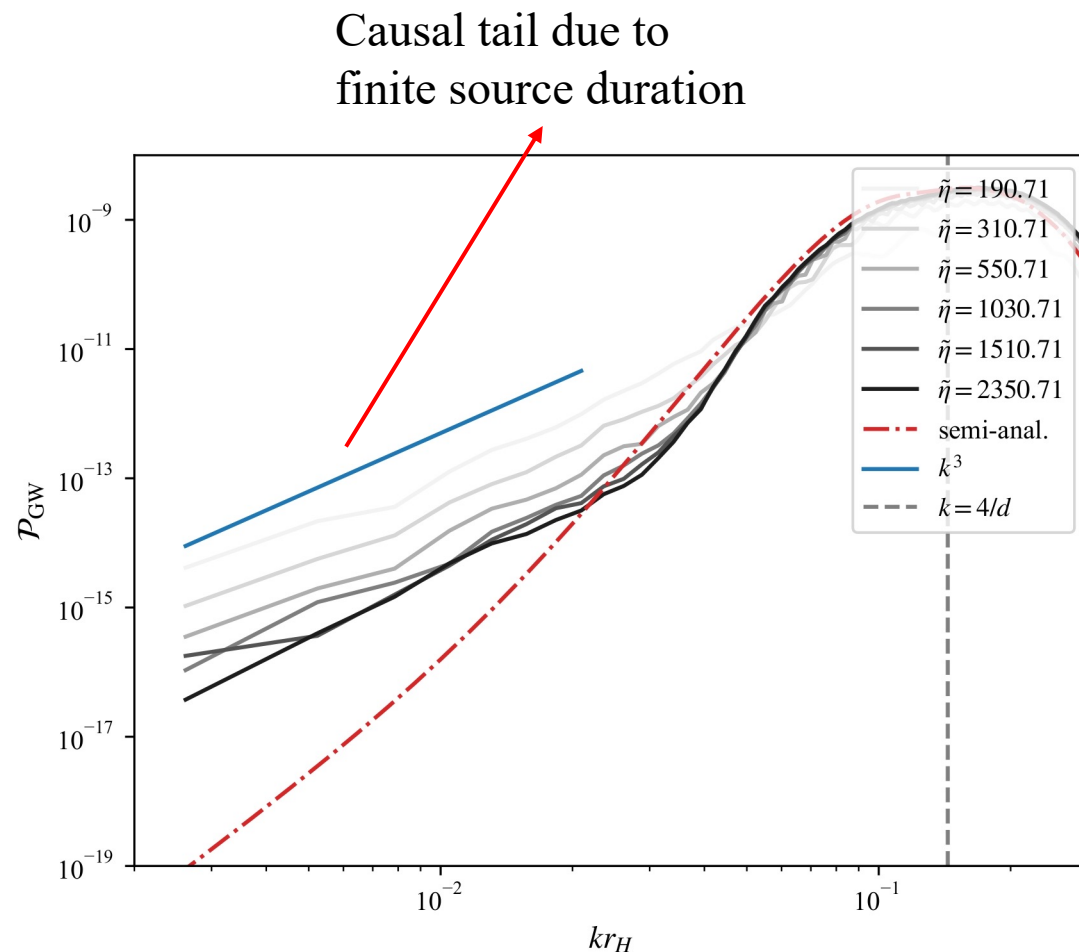
- Introduction
- Sound waves from collapsing curvature peaks
- Sound shell model
- 3D simulations of acoustic gravitational waves
- Numerical results**
- Conclusions and discussion

Shape of the acoustic GW energy spectrum

➤ $\mu = 0.4$



$\tilde{L} = 600, N_s = 50$



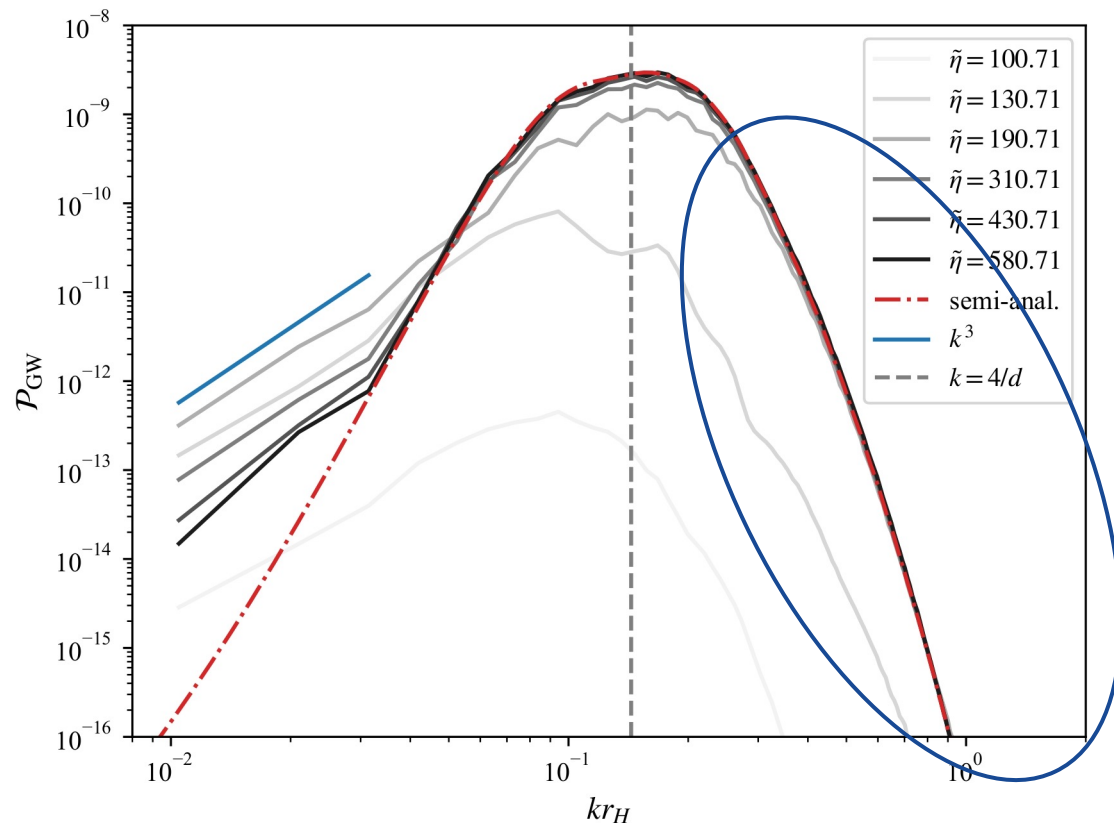
$\tilde{L} = 1200, N_s = 400$

Shape of the acoustic GW energy spectrum

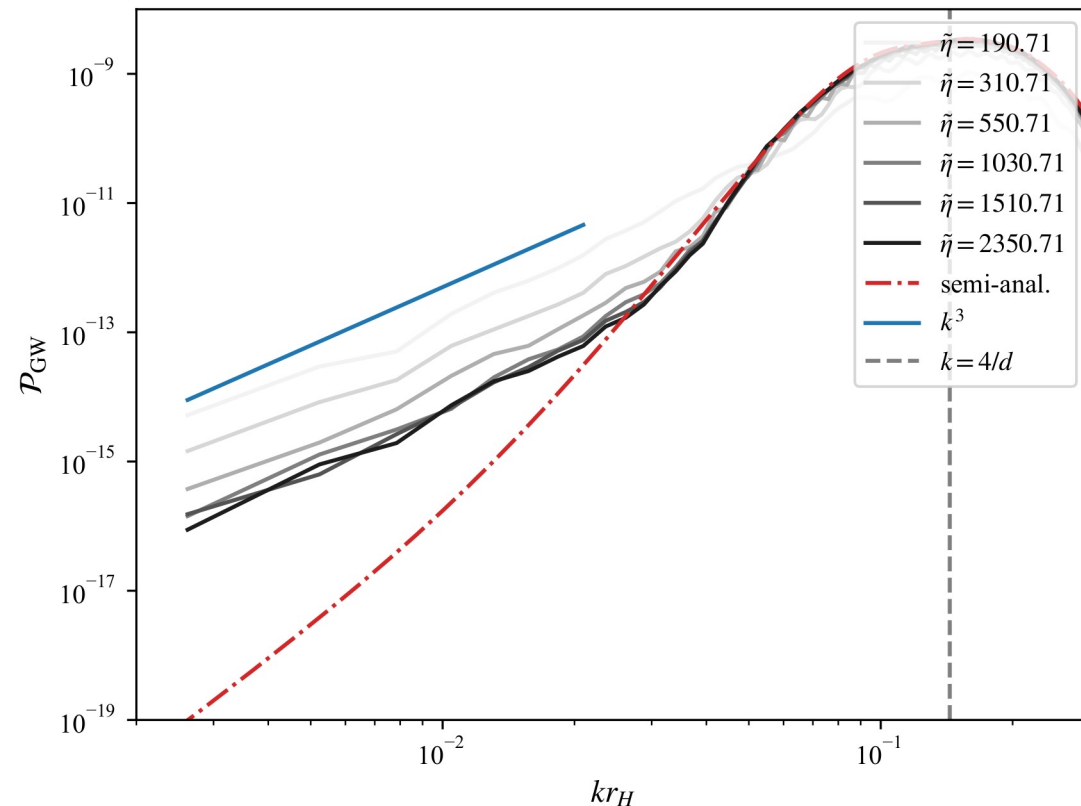
➤ $\mu = 0.4$ using the **linearized** hydrodynamical equations

$$3\delta' + 4\partial_i v^i = 0$$

$$4v^{i'} + \partial_i \delta = 0$$



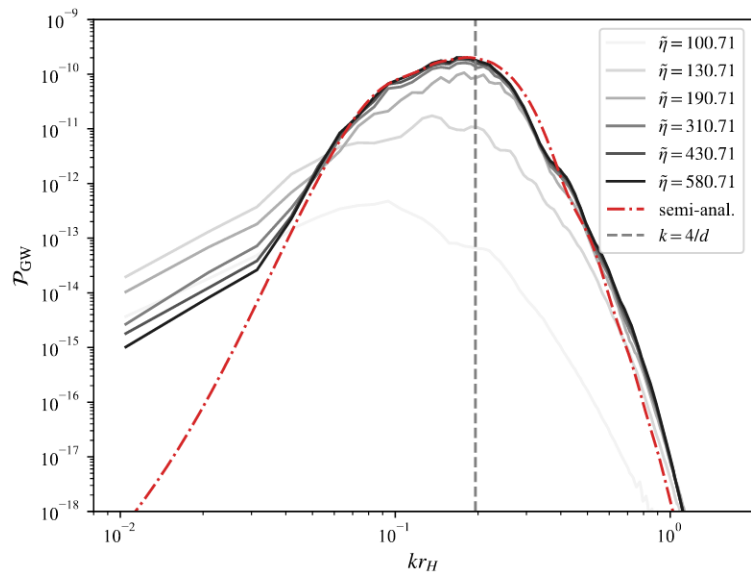
$\tilde{L} = 600, N_s = 50$



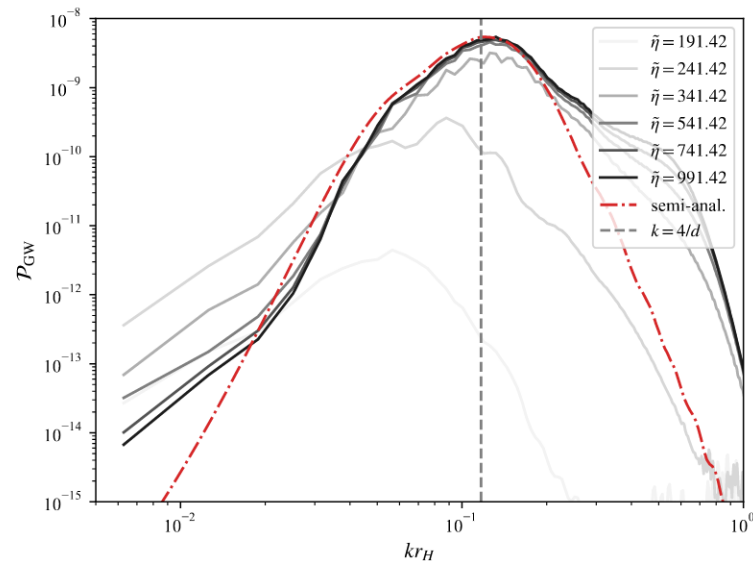
$\tilde{L} = 1200, N_s = 400$

Shape of the acoustic GW energy spectrum

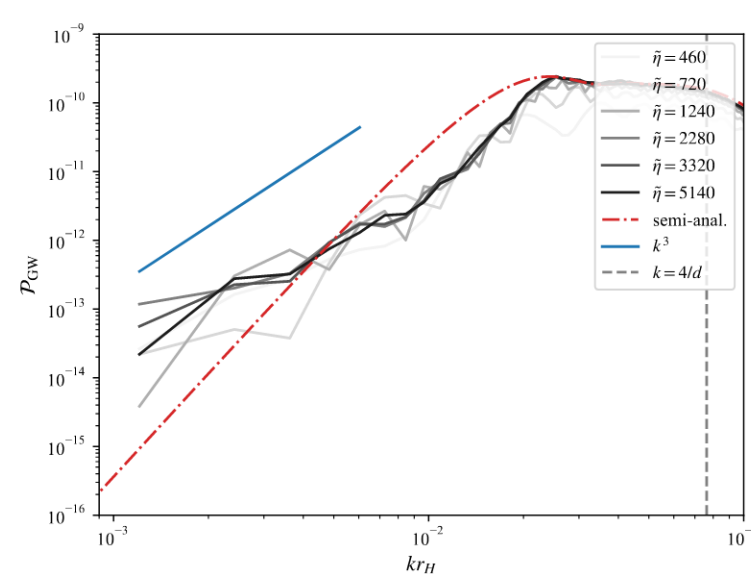
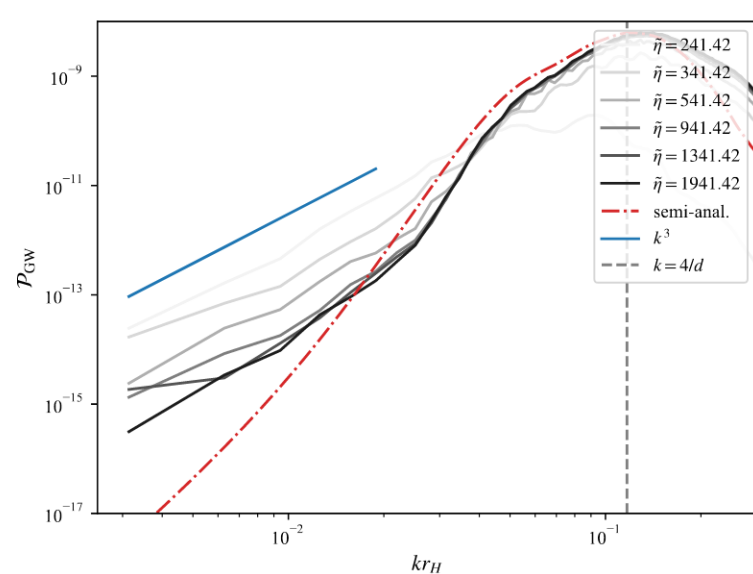
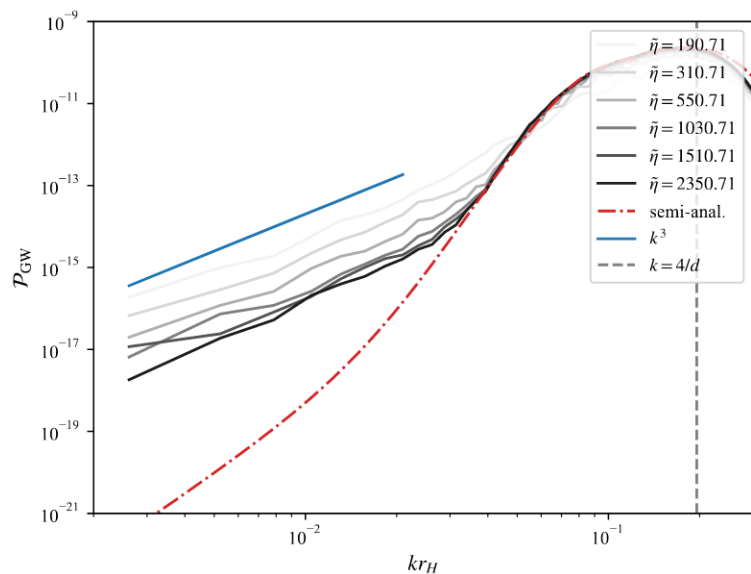
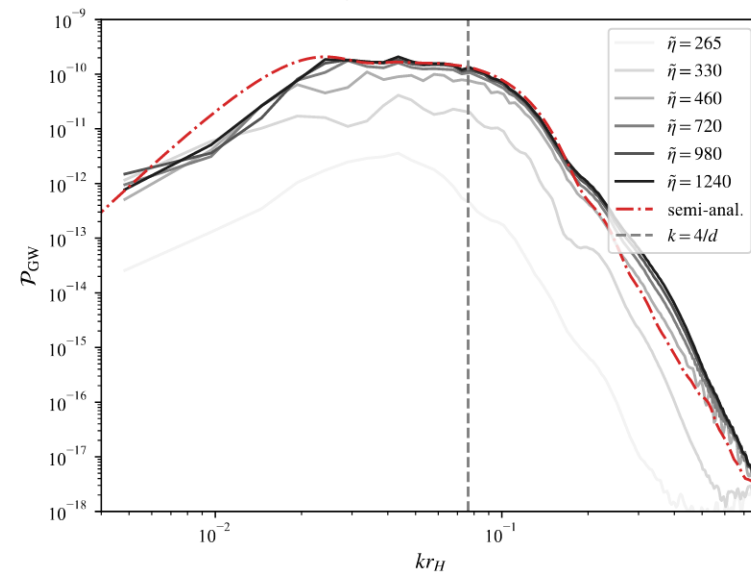
$\mu = -0.4$



$\mu = 0.8$

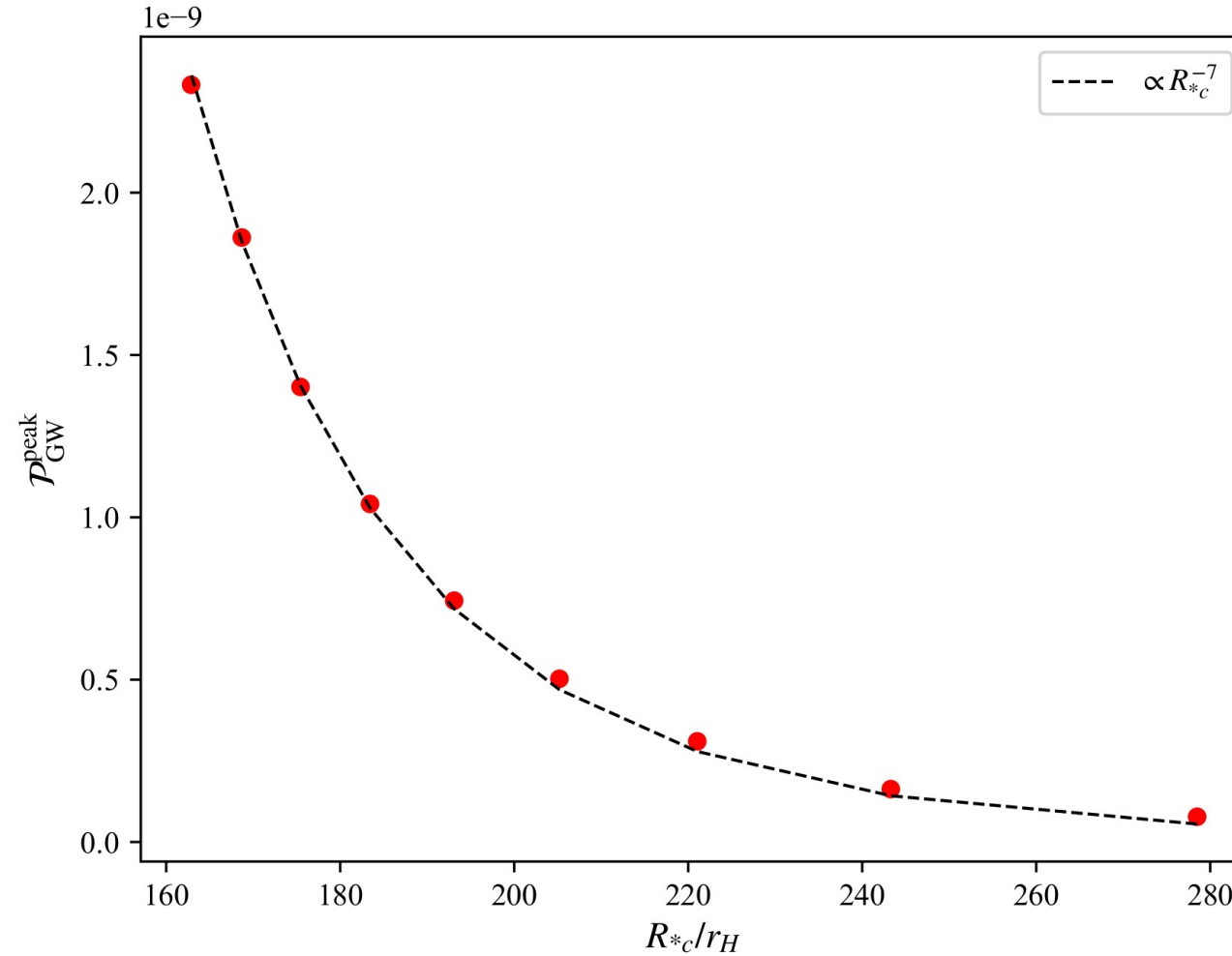


$\mu = 0.9$



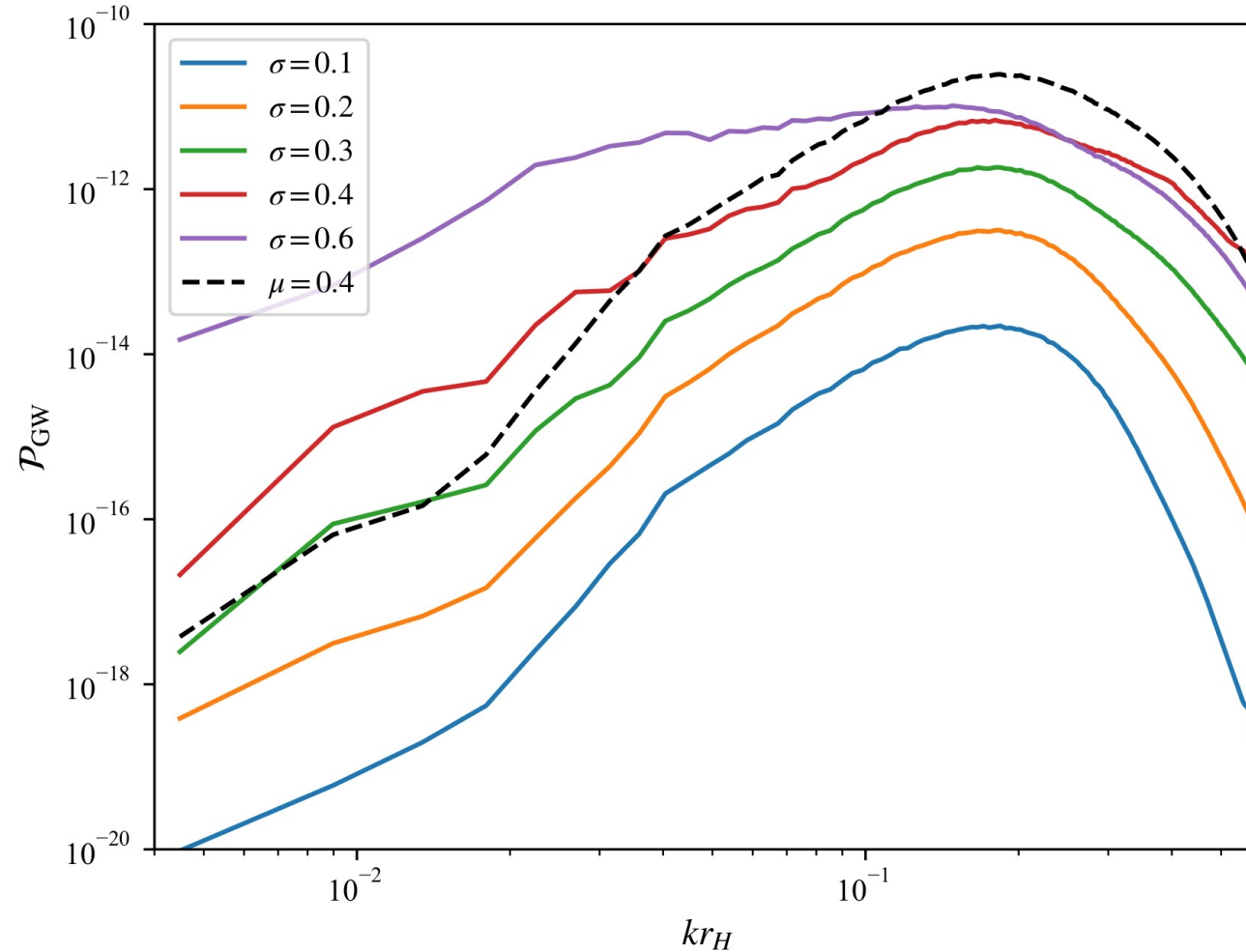
Effects of the mean comoving separations

➤ R_{*c}^{-7} scaling



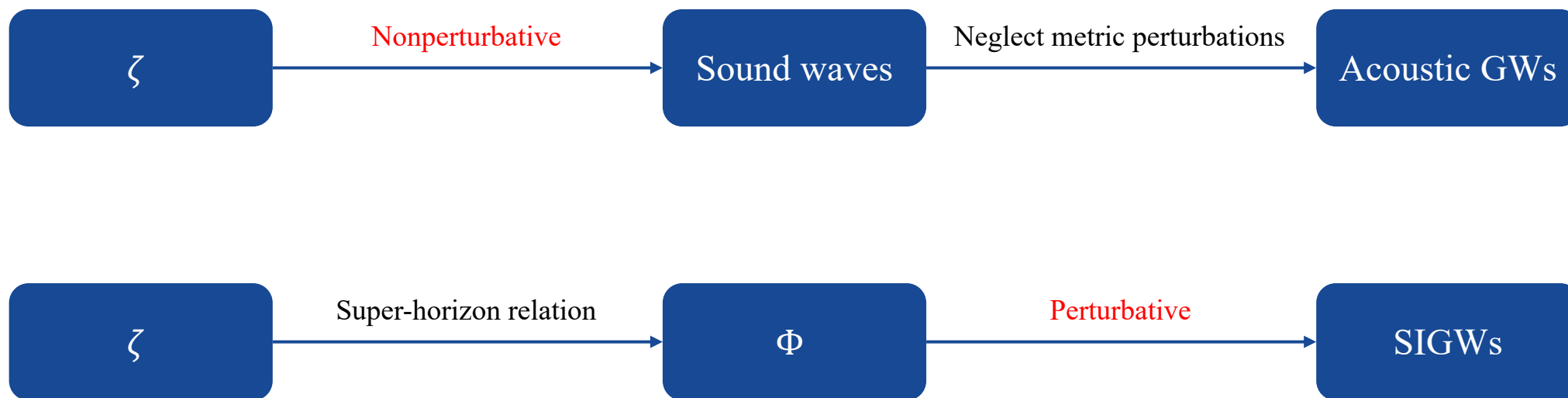
Effects of amplitude distributions of perturbations

- Gaussian probability density with **standard deviation** σ : $-1.28 \leq \mu \leq 1.28$



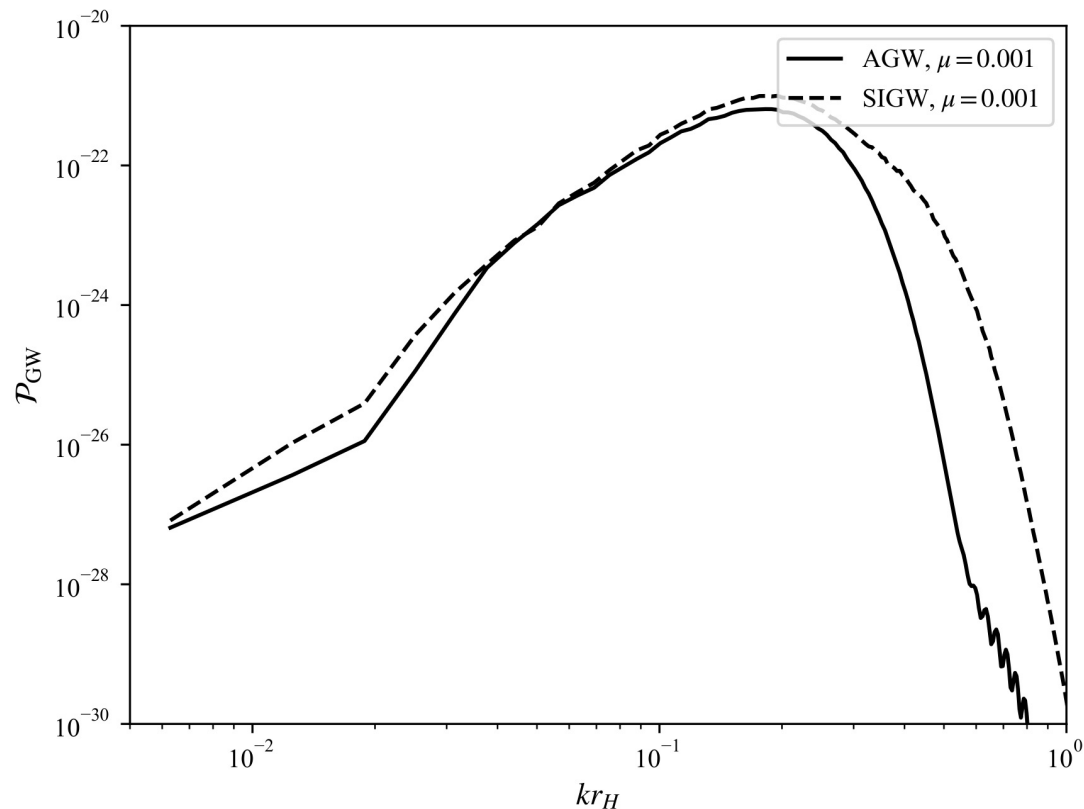
Comparisons with SIGWs

- From a **same real-space configuration** of curvature perturbations
- Solving SIGWs in real space [[Xiang-Xi Zeng, ZN, Rong-Gen Cai, Shao-Jiang Wang 2508.10812](#)]
- Sound wave configuration is the **fully GR evolutionary outcome** of the same curvature perturbations

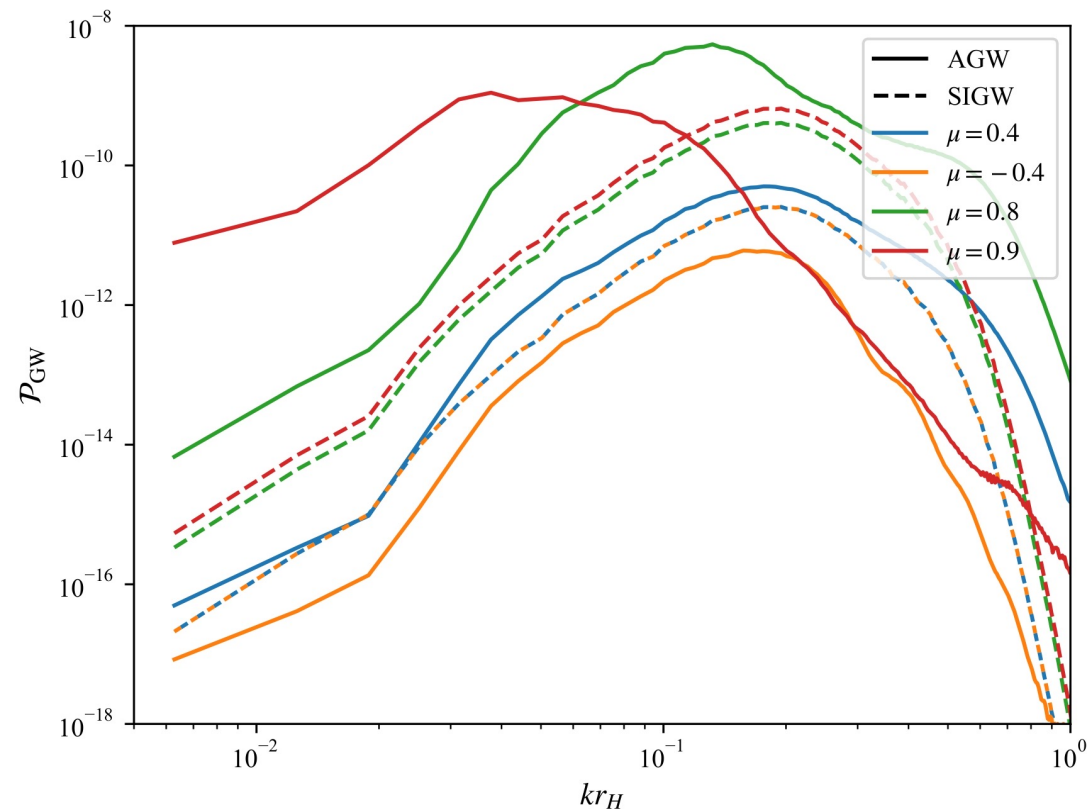


Comparisons with SIGWs

- SIGWs include additional contributions from **metric perturbations**, but **fluid motions** are dominant



Only slightly larger



Amplification of **an order of magnitude**, peak shift

Outline

- Introduction
- Sound waves from collapsing curvature peaks
- Sound shell model
- 3D simulations of acoustic gravitational waves
- Numerical results
- Conclusions and discussion**

Conclusions and discussion

- Acoustic GW peak frequency is controlled by the **shell thickness**
- Peak amplitude is extremely sensitive to R_{*c}
- **Causality** sets the IR scaling
- **Nonlinear hydrodynamics** enhances the UV power
- For small perturbations, it is simply the **pure fluid motion part** of SIGWs
- For large perturbations: **amplitude amplification and peak shift** due to **nonperturbative effects**
- More realistic treatment?
- End-to-end, fully GR simulations?
- Comparison with SIGWs from a common power spectrum?

Thank you for your attention!